EXERCISE 5

Higher-Order Inhomogeneous Equations with Constant Coefficients

Original statement · probls_ch3-ii_2627.pdf, p. 2

Third and Fourth Order • Repeated Roots • Undetermined Coefficients

STATEMENT

Working by analogy with the second order case, find the general solution of the following inhomogeneous higher-order linear equations with constant coefficients:

- (i) $\dfrac{d^3x}{dt^3} - 6\dfrac{d^2x}{dt^2} + 11\dfrac{dx}{dt} - 6x = e^{-t}$,
- (ii) $y''' - 3y' + 2 = \sin x$,
- (iii) $\dfrac{d^4x}{dt^4} - 4\dfrac{d^3x}{dt^3} + 8\dfrac{d^2x}{dt^2} - 8\dfrac{dx}{dt} + 4x = \sin t$.

SETUP, CLASSIFICATION AND A REMARK ON THE STATEMENT

All three are linear, constant-coefficient, inhomogeneous equations of order 3, 3 and 4. By the higher-order theory (Robinson Sec. 16.1, Topic 3 Sec. 3 box), the general solution of $a_n x^{(n)} + \cdots + a_0 x = f$ is $x = x_h + x_p$, where x_h contains n arbitrary constants (the solution space of the homogeneous equation has dimension n) and the method of undetermined coefficients gives x_p exactly as for $n = 2$. Hypotheses: coefficients constant, f continuous.

REMARK ON THE STATEMENT OF (II)

The sheet prints $y''' - 3y' + 2 = \sin x$. Read literally this is the linear equation $y''' - 3y' = \sin x - 2$ (the constant 2 moves to the right-hand side), which is perfectly solvable. The other two items and the form $L[y] = \ldots$ suggest that the term $2y$ may have been intended, but that is only a guess from the pattern of the sheet. We therefore solve **both readings in full**: $y''' - 3y' + 2y = \sin x$ (the guessed one) and the literal $y''' - 3y' = \sin x - 2$, each with its own general solution, so that the answer is complete whichever the author meant.

METHOD

Characteristic polynomial $P(k) = a_n k^n + \cdots + a_0$. A root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ of multiplicity m gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ for $j = 0, \ldots, m-1$ (Topic 3, Sec. 3 box, Robinson Sec. 16.1). The particular integral uses the guess table of Topic 3, Sec. 5, with the factor t^{m_s} when the exponent s of the forcing is a root of P of multiplicity m_s. The derivatives needed are now up to order n.

CLASS NOTES AND COMPLEMENT [NOTES, P. 6]

The notes state that the educated guess works 'the same for higher order'. The homogeneous part, with t^j factors for a root of multiplicity m, extends Cases 01 to 03 and is a complement (not in the class notes); the particular part uses the guess of the notes, times t^{m_s} when the exponent of the forcing is a root.

DERIVATION

(i) $x''' - 6x'' + 11x' - 6x = e^{-t}$

The homogeneous equation $\dddot{x} - 6\ddot{x} + 11\dot{x} - 6x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^3 - 6k^2 + 11k - 6 = 0$$

Factorising, $P(k) = (k-3)(k-2)(k-1)$, so the roots are

$$k = 1, \qquad k = 2, \qquad k = 3$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos \omega t$ and $t^j e^{\rho t}\sin \omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^t, e^{2t}, e^{3t}, so the complementary function is

$$x_h(t) = c_1 e^{t} + c_2 e^{2t} + c_3 e^{3t}$$

The forcing e^{-t} is an exponential with exponent $s = -1$; we test it in the characteristic polynomial: $P(-1) = -24$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 e^{-t}$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos \omega t$ and $\sin \omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 e^{-t} \\ x_p''(t) &= K_1 e^{-t} \\ x_p^{(3)}(t) &= -K_1 e^{-t} \end{aligned}$$

Substitute into $L[x_p] = x_p^{(3)} - 6x_p'' + 11x_p' - 6x_p$ and collect the terms that multiply the same function:

$$L[x_p] = -24K_1\, e^{-t}$$

This must equal the forcing $f(t) = e^{-t}$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$e^{-t}: \quad -24K_1 = 1$$

The linear system is

$$-24K_1 = 1 \quad (1)$$

$-24K_1 = 1$ contains one unknown only, so $K_1 = -\frac{1}{24}$.

Hence $K_1 = -\frac{1}{24}$.

Therefore

$$x_p(t) = -\frac{e^{-t}}{24}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^{t} + c_2 e^{2t} + c_3 e^{3t} - \frac{e^{-t}}{24}$$

(ii) $y''' - 3y' + 2y = \sin x$ **(independent variable** x**)**

The homogeneous equation $y^{(3)} - 3y' + 2y = 0$ has constant coefficients, so we try $y = e^{kx}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kx} \neq 0$ leaves the characteristic equation

$$P(k) = k^3 - 3k + 2 = 0$$

Factorising, $P(k) = (k-1)^2 (k+2)$, so the roots are

$$k = -2, \qquad k = 1 \text{ (multiplicity } 2)$$

A real root k of multiplicity m gives $x^j e^{kx}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $x^j e^{\rho x} \cos \omega x$ and $x^j e^{\rho x} \sin \omega x$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is e^{-2x}, e^x, xe^x, so the complementary function is

$$y_h(x) = c_1 e^{-2x} + c_2 e^x + c_3 x e^x$$

The forcing $\sin x$ is a sinusoidal term: it is the real or imaginary part of $e^{(i)x}$, so we test $s = i$ in the characteristic polynomial: $P(i) = 2 - 4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$y_p(x) = K_1 \cos x + K_2 \sin x$$

Differentiate term by term (product rule on $x^j e^{\sigma x}$, chain rule on $\cos \omega x$ and $\sin \omega x$):

$$\begin{aligned} y_p'(x) &= -K_1 \sin x + K_2 \cos x \\ y_p''(x) &= -K_1 \cos x - K_2 \sin x \\ y_p^{(3)}(x) &= K_1 \sin x - K_2 \cos x \end{aligned}$$

Substitute into $L[y_p] = y_p^{(3)} - 3y_p' + 2y_p$ and collect the terms that multiply the same function:

$$L[y_p] = (2K_1 - 4K_2) \cos x + (4K_1 + 2K_2) \sin x$$

This must equal the forcing $f(x) = \sin x$ for every x. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos x: &\quad 2K_1 - 4K_2 = 0 \\ \sin x: &\quad 4K_1 + 2K_2 = 1 \end{aligned}$$

The linear system is

$$\begin{aligned} 2K_1 - 4K_2 &= 0 \quad (1) \\ 4K_1 + 2K_2 &= 1 \quad (2) \end{aligned}$$

Solve $2K_1 - 4K_2 = 0$ for K_1: $K_1 = 2K_2$.

Substitute in $4K_1 + 2K_2 = 1$: $10K_2 = 1$.

$10K_2 = 1$ contains one unknown only, so $K_2 = \frac{1}{10}$.

Back-substitution: $K_1 = 2K_2 = \frac{1}{5}$.

Hence $K_1 = \frac{1}{5}$, $K_2 = \frac{1}{10}$.

Therefore

$$y_p(x) = \frac{\sin x}{10} + \frac{\cos x}{5}$$

The general solution is the complementary function plus the particular integral:

$$y(x) = y_h(x) + y_p(x) = c_1 e^{-2x} + c_2 e^{x} + c_3 x e^{x} + \frac{\sin x}{10} + \frac{\cos x}{5}$$

Literal reading of the sheet: $y''' - 3y' = \sin x - 2$

The homogeneous equation $y^{(3)} - 3y' = 0$ has constant coefficients, so we try $y = e^{kx}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kx} \neq 0$ leaves the characteristic equation

$$P(k) = k^3 - 3k = 0$$

Factorising, $P(k) = k\left(k^2 - 3\right)$, so the roots are

$$k = -\sqrt{3}, \qquad k = 0, \qquad k = \sqrt{3}$$

A real root k of multiplicity m gives $x^j e^{kx}$ for $j = 0, \ldots, m-1$; a complex pair $\rho \pm i\omega$ gives $x^j e^{\rho x} \cos \omega x$ and $x^j e^{\rho x} \sin \omega x$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^{-\sqrt{3}x}$, 1, $e^{\sqrt{3}x}$, so the complementary function is

$$y_h(x) = c_1 e^{-\sqrt{3}x} + c_2 + c_3 e^{\sqrt{3}x}$$

The forcing is a sum of terms. By linearity (Robinson Eq. 14.7) a particular integral is the sum of particular integrals of the separate terms, each found in turn.

Particular integral for the term $\sin x$

The forcing $\sin x$ is a sinusoidal term: it is the real or imaginary part of $e^{(i)x}$, so we test $s = i$ in the characteristic polynomial: $P(i) = -4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$y_p(x) = K_1 \cos x + K_2 \sin x$$

Differentiate term by term (product rule on $x^j e^{\sigma x}$, chain rule on $\cos \omega x$ and $\sin \omega x$):

$$\begin{aligned} y_p'(x) &= -K_1 \sin x + K_2 \cos x \\ y_p''(x) &= -K_1 \cos x - K_2 \sin x \\ y_p^{(3)}(x) &= K_1 \sin x - K_2 \cos x \end{aligned}$$

Substitute into $L[y_p] = y_p^{(3)} - 3y_p'$ and collect the terms that multiply the same function:

$$L[y_p] = -4K_2 \cos x + 4K_1 \sin x$$

This must equal the forcing $f(x) = \sin x$ for every x. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos x: \quad & -4K_2 = 0 \\ \sin x: \quad & 4K_1 = 1 \end{aligned}$$

The linear system is

$$\begin{aligned} -4K_2 &= 0 \quad (1) \\ 4K_1 &= 1 \quad (2) \end{aligned}$$

$-4K_2 = 0$ contains one unknown only, so $K_2 = 0$.

$4K_1 = 1$ contains one unknown only, so $K_1 = \frac{1}{4}$.

Hence $K_1 = \frac{1}{4}$, $K_2 = 0$.

Therefore

$$y_p(x) = \frac{\cos x}{4}$$

Particular integral for the term -2

The forcing -2 is an exponential with exponent $s = 0$; we test it in the characteristic polynomial: $P(0) = 0$.

Since $P(s) = 0$ and s is a root of multiplicity 1, the standard guess would contain terms of the complementary function. By the rule of thumb (Robinson Sec. 14.5, p. 139) the guess is multiplied by x^1.

Guess for the particular integral, with undetermined coefficients:

$$y_p(x) = K_1 x$$

Differentiate term by term (product rule on $x^j e^{\sigma x}$, chain rule on $\cos \omega x$ and $\sin \omega x$):

$$\begin{aligned} y_p'(x) &= K_1 \\ y_p''(x) &= 0 \\ y_p^{(3)}(x) &= 0 \end{aligned}$$

Substitute into $L[y_p] = y_p^{(3)} - 3y_p'$ and collect the terms that multiply the same function:

$$L[y_p] = -3K_1$$

This must equal the forcing $f(x) = -2$ for every x. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$1: \quad -3K_1 = -2$$

The linear system is

$$-3K_1 = -2 \quad (1)$$

$-3K_1 = -2$ contains one unknown only, so $K_1 = \frac{2}{3}$.

Hence $K_1 = \frac{2}{3}$.

Therefore

$$y_p(x) = \frac{2x}{3}$$

Adding the particular integrals,

$$y_p(x) = \frac{2x}{3} + \frac{\cos x}{4}$$

The general solution is the complementary function plus the particular integral:

$$y(x) = y_h(x) + y_p(x) = c_1 e^{-\sqrt{3}x} + c_2 + c_3 e^{\sqrt{3}x} + \frac{2x}{3} + \frac{\cos x}{4}$$

(iii) $x^{(4)} - 4x''' + 8x'' - 8x' + 4x = \sin t$

The homogeneous equation $x^{(4)} - 4\dddot{x} + 8\ddot{x} - 8\dot{x} + 4x = 0$ has constant coefficients, so we try $x = e^{kt}$ (Robinson Sec. 12.1 and 16.1): by the chain rule each derivative multiplies by k, and dividing by $e^{kt} \neq 0$ leaves the characteristic equation

$$P(k) = k^4 - 4k^3 + 8k^2 - 8k + 4 = 0$$

Factorising, $P(k) = (k - 1 - i)^2 (k - 1 + i)^2$, so the roots are

$$k = 1 - i \text{ (multiplicity 2)}, \qquad k = 1 + i \text{ (multiplicity 2)}$$

A real root k of multiplicity m gives $t^j e^{kt}$ for $j = 0, \dots, m-1$; a complex pair $\rho \pm i\omega$ gives $t^j e^{\rho t}\cos\omega t$ and $t^j e^{\rho t}\sin\omega t$ (Euler's formula, Robinson Sec. 12.3). The fundamental set is $e^t\cos t$, $e^t \sin t$, $te^t\cos t$, $te^t \sin t$, so the complementary function is

$$x_h(t) = c_1 e^t \cos t + c_2 e^t \sin t + c_3 t e^t \cos t + c_4 t e^t \sin t$$

The forcing $\sin t$ is a sinusoidal term: it is the real or imaginary part of $e^{(i)t}$, so we test $s = i$ in the characteristic polynomial: $P(i) = -3 - 4i$.

Since $P(s) \neq 0$, no term of the standard guess solves the homogeneous equation, so the standard guess is used as it stands.

Guess for the particular integral, with undetermined coefficients:

$$x_p(t) = K_1 \cos t + K_2 \sin t$$

Differentiate term by term (product rule on $t^j e^{\sigma t}$, chain rule on $\cos\omega t$ and $\sin\omega t$):

$$\begin{aligned} x_p'(t) &= -K_1 \sin t + K_2 \cos t \\ x_p''(t) &= -K_1 \cos t - K_2 \sin t \\ x_p^{(3)}(t) &= K_1 \sin t - K_2 \cos t \\ x_p^{(4)}(t) &= K_1 \cos t + K_2 \sin t \end{aligned}$$

Substitute into $L[x_p] = x_p^{(4)} - 4x_p^{(3)} + 8x_p'' - 8x_p' + 4x_p$ and collect the terms that multiply the same function:

$$L[x_p] = (-3K_1 - 4K_2)\,\cos t + (4K_1 - 3K_2)\,\sin t$$

This must equal the forcing $f(t) = \sin t$ for every t. The functions on the left are linearly independent, so the coefficient of each must match the coefficient of the same function in f:

$$\begin{aligned} \cos t: &\quad -3K_1 - 4K_2 = 0 \\ \sin t: &\quad 4K_1 - 3K_2 = 1 \end{aligned}$$

The linear system is

$$-3K_1 - 4K_2 = 0 \quad (1)$$
$$4K_1 - 3K_2 = 1 \quad (2)$$

Solve $-3K_1 - 4K_2 = 0$ for K_1: $K_1 = -\frac{4K_2}{3}$.

Substitute in $4K_1 - 3K_2 = 1$: $-\frac{25K_2}{3} = 1$.

$-\frac{25K_2}{3} = 1$ contains one unknown only, so $K_2 = -\frac{3}{25}$.

Back-substitution: $K_1 = -\frac{4K_2}{3} = \frac{4}{25}$.

Hence $K_1 = \frac{4}{25}$, $K_2 = -\frac{3}{25}$.

Therefore

$$x_p(t) = -\frac{3\sin t}{25} + \frac{4\cos t}{25}$$

The general solution is the complementary function plus the particular integral:

$$x(t) = x_h(t) + x_p(t) = c_1 e^t \cos t + c_2 e^t \sin t + c_3 t e^t \cos t + c_4 t e^t \sin t - \frac{3\sin t}{25} + \frac{4\cos t}{25}$$

INTERPRETATION AND CHECKS

Item	General solution
(i)	$c_1 e^t + c_2 e^{2t} + c_3 e^{3t} - \frac{e^{-t}}{24}$
(ii), intended	$y(x) = c_1 e^{-2x} + c_2 e^x + c_3 x e^x + \frac{\sin x}{10} + \frac{\cos x}{5}$
(ii), literal reading	$y(x) = c_1 e^{-\sqrt{3}x} + c_2 + c_3 e^{\sqrt{3}x} + \frac{2x}{3} + \frac{\cos x}{4}$
(iii)	$c_1 e^t \cos t + c_2 e^t \sin t + c_3 t e^t \cos t + c_4 t e^t \sin t - \frac{3\sin t}{25} + \frac{4\cos t}{25}$

- The number of arbitrary constants equals the order (3, 3, 4), as the dimension of the solution space requires (Robinson Sec. 16.2).
- (i): three distinct real roots $1, 2, 3$, none equal to the exponent $s = -1$ of the forcing, hence no factor t.
- (ii): $P(k) = k^3 - 3k + 2 = (k-1)^2(k+2)$ has the **double** root 1, giving e^x and xe^x; the forcing exponent is $s = \pm i$, not a root.
- (iii): $P(k) = k^4 - 4k^3 + 8k^2 - 8k + 4 = (k^2 - 2k + 2)^2 = ((k-1)^2 + 1)^2$ has the **double** complex pair $1 \pm i$, giving $e^t \cos t$, $e^t \sin t$, $te^t \cos t$, $te^t \sin t$; the forcing exponent is $\pm i$, not a root.

- Each particular integral was substituted back into its equation symbolically (script `scripts/verify_topic3_topic4.py`); the residual vanishes identically.